IronMind AI — Presentation & Viva Survival Guide

Quick Reference Cheat Sheet for Presentation & Project Evaluation

Prepared for Academic Defense • Live Mobile Demonstration • Production Release v1.0.0

🎯 Part 1: Your 30-Second Opening Speech

When the evaluators ask: "*Please introduce your project*", deliver this exact speech:

> *"Good morning/afternoon, Sir/Madam.*
>
> *Our project is **IronMind AI** — an on-device, real-time AI biomechanical fitness coach and posture analysis platform.*
>
> *When individuals exercise without supervision, improper technique leads to severe joint strain and musculoskeletal injuries. Personal trainers are expensive, and wearable sensors are uncomfortable. IronMind AI solves this by turning any standard smartphone camera into an intelligent posture analyzer.*
>
> *Using **Edge Computer Vision (Google BlazePose)**, the app tracks **33 anatomical 3D joints** at 30+ FPS, calculates dynamic joint angles using trigonometry, filters optical noise with an EMA smoothing filter, counts repetitions via finite state machines, and provides instant audio-visual coaching cues — all while preserving 100% user privacy by keeping video processing entirely on the phone."*

🧠 Part 2: How It Works in 3 Simple Steps

1. The Eyes (Vision): The smartphone camera captures video frames. The on-device BlazePose neural network predicts 33 3D coordinate keypoints (X, Y, Z) across the body in sub-30ms without sending any video to the internet.

2. The Brain (Math & Kinematics): The app calculates the exact interior angle between joints using `atan2` trigonometry (e.g. Hip → Knee → Ankle angle during squats). An **Exponential Moving Average (EMA)** filter ($\alpha = 0.18$) removes camera shake so optical jitter never causes false rep counts.

3. The Coach (State Machine & Feedback): A multi-phase state machine tracks movement (e.g., *Standing → Descending → Bottom Hold → Ascending*). When full depth is reached with good spinal alignment, a repetition increments, and a Live Form Score (e.g., 94%) is displayed.

🛠️ Part 3: Technology Stack Summary

Component	Technology	Simple Explanation
Frontend (Mobile App)	Flutter & Dart 3.x	Cross-platform UI framework with 60 FPS performance and dark Material 3 design.
Edge AI Model	Google ML Kit (BlazePose)	Lightweight deep learning model trained on COCO and GHUM 3D datasets for 33 3D landmarks.
Kinematic Engine	Custom Trigonometry & EMA	Mathematical engine for atan2 joint angles, noise smoothing, and baseline auto-calibration.
Backend API	Python 3.12 & FastAPI	Asynchronous REST API server handling user authentication, statistics, and workout sync.
Database & Storage	SQLite 3 + SharedPreferences	**Offline-First:** Works without internet via phone storage, syncs to SQLite when online.
Security	PBKDF2-HMAC-SHA256	100,000 hashing rounds + 16-byte random salt to protect user passwords.

📱 Part 4: Step-by-Step Live Demo Script

Follow these exact steps when demonstrating the app on your phone:

Step 1: Open the App & Sign In
Open IronMind AI on the phone. Tap *"I already have an account"* and enter the demo credentials:
• **Email:** `demo@ironmind.ai` • **Password:** `password123`
(Point out to evaluators: "Notice how quickly authentication authenticates and loads user workout profiles.")

Step 2: Show the Dashboard
Show the clean dark UI with 5 exercises: **Squats, Push-ups, Lunges, Jumping Jacks, and Plank**.
(Point out: "The dashboard displays daily streaks, targeted fitness goals, and personalized exercise recommendations.")

Step 3: Start AI Camera Session (Squats)
Tap **Squats** → tap **Start Camera Session**. Step back 6–8 feet so your full body is visible.
(Point out: "The BlazePose model immediately tracks 33 landmarks and performs initial standing calibration.")

Step 4: Perform 2 to 3 Squats
Perform a squat. Show the screen to the evaluators:
• The **Rep Counter** increments: `0 → 1 → 2`.
• The **Form Score** displays in real time: `94% - Excellent Form`.
• The **Live Coach Cue** shows: `"Good depth! Keep back straight."`

Step 5: Finish & Show Analytics
Tap the checkmark button to finish workout. Show the **Workout Summary** and **Analytics Tab** with historical progress charts and total volume.

🎓 Part 5: Top 8 Tough Viva Questions & Winning Answers

Q1: Why did you run the AI on the smartphone rather than a powerful cloud server?

Answer: Cloud streaming 30 FPS video introduces 150–300ms latency, which breaks real-time coaching cues. On-device inference delivers sub-30ms response times, works 100% offline, and ensures complete user privacy because raw camera frames never leave the phone.

Q2: What datasets were used to train the pose model?

Answer: The BlazePose model was trained on the **COCO Keypoints Dataset** (over 200k diverse real-world images for 2D keypoints) and the **Google GHUM 3D Dataset** (a generative 3D human shape dataset providing 3D depth and occlusion handling).

Q3: How does the system prevent camera shake from counting false reps?

Answer: We apply an **Exponential Moving Average (EMA) filter** ($\alpha = 0.18$) on calculated joint angles to filter optical jitter, combined with a 4-phase finite state machine that requires full range of motion before incrementing a rep.

Q4: How do you calculate joint angles in 3D?

Answer: Using the 2-argument arctangent formula `atan2(Δy, Δx)` between three consecutive anatomical landmarks (such as Hip, Knee, and Ankle), normalized between 0° and 180°.

Q5: What happens if there is no internet connection during a workout?

Answer: The app utilizes an **offline-first architecture**. It saves user sessions into Android `SharedPreferences` locally and synchronizes with the SQLite backend once network connectivity is re-established.

Q6: How do you adapt to different people's heights and limb lengths?

Answer: We implement an **auto-calibration baseline phase** that captures $N=30$ standing frames at the start of each session to dynamically adjust depth thresholds to individual body proportions.

Q7: How is user password security handled?

Answer: Passwords are never stored in plaintext. They are salted with a 16-byte cryptographic random salt and hashed using **PBKDF2-HMAC-SHA256 with 100,000 iterations**.

Q8: Can this app be deployed on other platforms?

Answer: Yes, we built a standalone 64-bit release APK (`app-release.apk`) that installs on any modern Android device, and also developed a pure Python (OpenCV + MediaPipe) desktop research edition.
